CAUSE & EFFECT MATRIX — LCP-01 Scrubber & Burner Control

Project: Burner PetroBaker | Panel: LCP-01 | Ref. drawing: PH-LCP-01 V1 | Document: PH-LCP-01-CE | Rev. A (for client review) | 26-10-2026

Purpose	Defines the PLC (LCP-01) response to every process and safety cause, as requested in the client e-mail of 04-Oct-2026 (Eng. Amr Farag).
Scope	Gas scrubber alarms/trips, ESD, burner BR01 (Riello RS100) and BR02 (DLX 53 SHFP) trips and permissives, tank temperature control (client sheet 'Temperature control of burner'). This matrix lists causes and effects only. Operating modes (AUTO / MAN), START / STOP, RESET, start-up after power loss and lamps are described in the Control Narrative (docs/التشغيل.md).
Not in scope (Rev. A)	Communication with the Master PLC in the control room — deferred to a later revision. Scrubber low level switch — none on this scrubber (DI I12.5 spare).
Burner flame safety	Purge, ignition, flame supervision and the pilot / main gas valves are handled by each burner's own certified control box (RS100: RMG88; DLX 53: SQ DUAL). LCP-01 only gives the burner its heat request (TL / L) and stage request (TR / T2), and reads its RUN and LOCKOUT signals. LCP-01 never drives a gas valve directly.
Hardwired safety	E-STOP S0 opens the 230 V control supply of BR01 (NC2) and BR02 (NC3) directly, independent of the PLC. Safety thermostat TS (manual reset) is in series with the BR01 control supply. These act even if the PLC fails.
Fail-safe outputs	All PLC outputs de-energise on PLC STOP or 24 VDC loss. Valve types were chosen so that de-energised = safe: XV-401 flare (NO) OPENS, XV-402 drain (NC) CLOSES, XV-403 suction (NC) CLOSES, both burners lose their heat request. The drain is the only valve that is energised to act: it opens on high level (C04 / C05).
How to read the matrix	One row per cause. Cells under each effect say what happens to that output while the cause is active. Empty = no action. STOP = output de-energised / burner request removed. RUN = energised. OPEN / CLOSE = resulting valve position. ALARM = HMI alarm (the burner fault lamps H3 / H4 are wired to the burner lockout, not to the PLC). Type: Trip = stops and stays stopped until RESET (S2). Permissive = stops, restarts by itself when the condition clears (AUTO mode, burner enabled). Control = normal temperature regulation. Alarm = HMI message only. Inherent = happens by wiring, no program. Column 'Source': Client = client document; PH decision = decided by PH (Eng. Ahmed Zain); PH proposal = proposed by PH, still open.
Requested from client	On sheet 'C&E Matrix', fill the two yellow columns for every row: 'Client response' (Approved / Approved with comment / Rejected) and 'Client comment'. Rows marked 'PH proposal' need particular attention. Use the filter on the header row, e.g. Source = PH proposal. Parameter values are on sheet 'Parameters' and are adjustable from the HMI.

Revision	Date	Description	By / Approved
A	26-10-2026	First issue for client review	PH / Eng. Ahmed Zain

CAUSE & EFFECT MATRIX

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No.	Tag	Cause	PLC address	Condition / setpoint	Type	EFFECTS								Reset	Source	Remarks	Client response	Client comment
						E1 BR01 heat request (TL)	E2 BR01 2nd stage (TR)	E3 BR02 start (L)	E4 BR02 high fire (T2)	E5 XV-401 Flare SV (NO)	E6 XV-402 Drain SV (NC)	E7 XV-403 Suction SV (NC)	E8 HMI alarm					
						Q0.0 / R1	Q0.1 / R2	Q0.5 / R6	Q0.6 / R7	Q0.2 / R3	Q0.3 / R4	Q0.4 / R5	HMI					
EMERGENCY SHUTDOWN																		
C01	S0 (E-STOP)	E-STOP pressed	I1.0	NC1 contact open	Trip	**STOP**	**STOP**	**STOP**	**STOP**	**OPEN**	**CLOSE**	**CLOSE**	**ALARM**	Release S0 (E-STOP), then S2 (RESET); burners need START (S1 / S6)	PH proposal	Burners are also cut by hardwired S0 NC2/NC3 (client e-mail: ESD impact on burner ON/OFF). Scrubber valves to safe position on ESD = PH proposal. The drain stays closed during E-STOP even on high level (meeting 02-Nov-2026).		
C02	—	PLC STOP, 24 VDC loss or power-up	—	All outputs de-energised	Inherent	**STOP**	**STOP**	**STOP**	**STOP**	**OPEN**	**CLOSE**	**CLOSE**		S2 (RESET) for scrubber; burners need START (S1 / S6)	PH proposal	Result of valve types (flare NO / drain NC / suction NC) and relay outputs. After power returns flare and suction stay in trip until S2. The drain is not held: it follows the level as soon as the PLC runs (C04), otherwise a high level at power-up could never be cleared. Drain type NC confirmed 02-Nov-2026; drawing V1 shows NO and will be revised (Eng. Omar).		
GAS SCRUBBER																		
C03	PSH-101	Scrubber high pressure	I0.0	Contact open (fail-safe)	Trip					**OPEN**		**CLOSE**	**ALARM**	S2 (RESET) after cleared	PH decision	Client e-mail asks for scrubber alarms/trips; the valve action and burner policy were decided by PH. Burners are NOT tripped (decision PH / Eng. Ahmed Zain); if gas pressure to the burner drops, the burner's own gas pressure switch stops it (C07g / C08g). The drain is not touched by PSH; it follows the level (meeting 02-Nov-2026).		
C04	LSH-102	Scrubber high level	I0.1	Contact open (fail-safe) for TDRN_ON (3 s)	Control						**OPEN**		**ALARM**	Automatic: drain closes when LSH-102 healthy for TDRN_OFF (5 s)	PH decision	Client e-mail asks for the alarm. Opening the drain on LSH-102 was decided in the meeting of 02-Nov-2026: the liquid drains a little and the level falls. Also opened by LSHH-103 (C05). Closes again above the drain nozzle, so liquid keeps sealing the drain. Not during E-STOP.		
C04a	XV-402	Drain open too long (LSH-102 stuck or wire broken)	—	Drain open continuously for TDRN_MAX (10 min)	Trip						**CLOSE**		**ALARM**	S2 (RESET); the drain then follows the level again	PH decision	The scrubber has no low level switch (C06). A broken LSH-102 wire reads as high level and would keep the drain open until the scrubber is empty and gas blows through the drain. This timer is the only protection against that. Accepted 02-Nov-2026.		
C04b	LSH-102 / LSHH-103	LSH-102 fault: high-high level without high level	—	LSHH-103 open while LSH-102 healthy, for 5 s	Alarm								**ALARM**	S2 (RESET) when the two switches agree	PH decision	LSHH-103 is mounted above LSH-102, so it cannot trip alone unless LSH-102 has failed. Accepted 02-Nov-2026.		
C05	LSHH-103	Scrubber high-high level	I0.2	Contact open (fail-safe)	Trip						**OPEN**	**CLOSE**	**ALARM**	S2 (RESET) after cleared (suction)	PH decision	Burners are NOT tripped (as C03). Flare not touched (meeting 02-Nov-2026). The drain opens as for C04 and is not held by the trip: it follows the level, so it still opens if LSH-102 has failed.		
C06	LSL (scrubber)	Scrubber low level	I12.5 (spare)	—	Not installed									—	PH decision	Not installed: the scrubber has LSH-102 and LSHH-103 only (confirmed 02-Nov-2026). I12.5 is spare.		
C06a	S0 / PSH-101 / LSHH-103 / power-up	Scrubber trip cause memory and first-out	—	EStopOK=0 / PSH open / LSHH open / FirstScan	Alarm								**ALARM**	S2 (RESET) with PSH, LSHH and EStopOK healthy	PH proposal	F44 / 0007 D5: each cause is latched until RESET, including causes arriving after the first. FirstOut: 0=none, 1=E-STOP, 2=PSH-101, 3=LSHH-103, 4=power-up. First cause holds until RESET; simultaneous priority: E-STOP, PSH, LSHH, power-up. Capture power-up inside Scrubber before Main clears FirstScan. HMI indication only; existing valve/burner policy unchanged. Accepted internally by user 2026-11-01; remains PH proposal, not a Client requirement.		
BURNER BR01 — tank TK01 (operating modes, START / STOP: see Control Narrative)																		
C07a	LSL-TK01	TK01 level below low level switch	I1.2	LSL not reached (contact open)	Permissive	**STOP**	**STOP**						**ALARM**	Automatic when level returns	Client	Client sheet: burner may run only when level has reached LSL. Contact closes with liquid present, so a broken wire also blocks the burner. PH proposal (0007): HMI warning 'TK01 level low: burner held'; clears automatically when level returns.		
C07b	TE-501 / TE-502	TK01 oil temperature at setpoint	IW96 / IW98	max(T) >= SP (70 °C)	Control	**STOP**	**STOP**							Automatic at max(T) <= SP − HYST (65 °C)	Client	Client cases 2/3/6. The higher of inlet/outlet is used and the 5 °C restart band are PH proposals: each restart costs ~50 s of purge.		
C07c	TE-501 / TE-502	TK01 oil temperature low — fast heat-up	IW96 / IW98	max(T) < SP − STG (60 °C)	Control		**RUN**							Automatic at max(T) >= 62 °C (2 °C band)	PH proposal	2nd stage / high fire. Between 60 and 70 °C the burner runs on 1st stage. Back to 1st stage at 62 °C: the 2 °C band stops relay R2 / R7 chattering at 60 °C.		
C07d	TE-501 / TE-502	Thermocouple fault — either of the two	IW96 / IW98	Either channel: 32767 / −32768, outside −20…150 °C, or channel diagnostic	Trip	**STOP**	**STOP**						**ALARM**	S2 (RESET) after fault cleared	PH proposal	Each channel is checked BEFORE taking the higher value, so one failed-low sensor cannot hide behind the other. Heat is inhibited until both channels are valid after power-up.		
C07e	Rflt (BR01 fault relay)	BR01 lockout (flame / ignition / air pressure failure)	I0.5	Lockout signal present	Trip	**STOP**	**STOP**						**ALARM**	At burner control box, then S2 (RESET)	PH proposal	The burner's own control box has already closed the gas valves. PLC removes the heat request so the burner does not restart by itself after a local reset.		
C07f	RR (BR01 run relay)	BR01 failed to start	I0.3	Heat request ON, no RUN within T_START (120 s)	Trip	**STOP**	**STOP**						**ALARM**	S2 (RESET)	PH proposal	RUN = gas valves energised (firing). Normal start ~55 s (RS100 §6.8.1) / ~65 s (DLX 53).		
C07g	RR (BR01 run relay)	BR01 stopped by itself while running	I0.3	RUN lost, no lockout, heat request still ON	Trip	**STOP**	**STOP**						**ALARM**	S2 (RESET)	PH proposal	E.g. safety thermostat TS, gas pressure switch, burner supply lost. Without this row the PLC would keep requesting heat from a burner that is no longer firing.		
C07h	—	BR01 running continuously for 24 h	—	Run timer >= 24 h	Permissive	**STOP**	**STOP**							Automatic (restarts on next demand)	PH proposal	RS100 is approved for intermittent operation and must stop at least once every 24 h (RS100 manual §5.10).		
BURNER BR02 — tank TK02 (operating modes, START / STOP: see Control Narrative)																		
C08a	LSL-TK02	TK02 level below low level switch	I1.3	LSL not reached (contact open)	Permissive			**STOP**	**STOP**				**ALARM**	Automatic when level returns	Client	Client sheet: burner may run only when level has reached LSL. Contact closes with liquid present, so a broken wire also blocks the burner. PH proposal (0007): HMI warning 'TK02 level low: burner held'; clears automatically when level returns.		
C08b	TE-503 / TE-504	TK02 oil temperature at setpoint	IW100 / IW102	max(T) >= SP (70 °C)	Control			**STOP**	**STOP**					Automatic at max(T) <= SP − HYST (65 °C)	Client	Client cases 2/3/6. The higher of inlet/outlet is used and the 5 °C restart band are PH proposals: each restart costs ~50 s of purge.		
C08c	TE-503 / TE-504	TK02 oil temperature low — fast heat-up	IW100 / IW102	max(T) < SP − STG (60 °C)	Control				**RUN**					Automatic at max(T) >= 62 °C (2 °C band)	PH proposal	2nd stage / high fire. Between 60 and 70 °C the burner runs on 1st stage. Back to 1st stage at 62 °C: the 2 °C band stops relay R2 / R7 chattering at 60 °C.		
C08d	TE-503 / TE-504	Thermocouple fault — either of the two	IW100 / IW102	Either channel: 32767 / −32768, outside −20…150 °C, or channel diagnostic	Trip			**STOP**	**STOP**				**ALARM**	S2 (RESET) after fault cleared	PH proposal	Each channel is checked BEFORE taking the higher value, so one failed-low sensor cannot hide behind the other. Heat is inhibited until both channels are valid after power-up.		
C08e	Rflt2 (BR02 fault relay)	BR02 lockout (flame / ignition / air pressure failure)	I1.5	Lockout signal present	Trip			**STOP**	**STOP**				**ALARM**	At burner control box, then S2 (RESET)	PH proposal	The burner's own control box has already closed the gas valves. PLC removes the heat request so the burner does not restart by itself after a local reset.		
C08f	RR2 (BR02 run relay)	BR02 failed to start	I1.4	Heat request ON, no RUN within T_START (120 s)	Trip			**STOP**	**STOP**				**ALARM**	S2 (RESET)	PH proposal	RUN = gas valves energised (firing). Normal start ~55 s (RS100 §6.8.1) / ~65 s (DLX 53).		
C08g	RR2 (BR02 run relay)	BR02 stopped by itself while running	I1.4	RUN lost, no lockout, heat request still ON	Trip			**STOP**	**STOP**				**ALARM**	S2 (RESET)	PH proposal	E.g. safety thermostat TS, gas pressure switch, burner supply lost. Without this row the PLC would keep requesting heat from a burner that is no longer firing.		

ADJUSTABLE PARAMETERS (HMI)

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Parameter	Description	Default	Unit	Range	Source
SP	Tank temperature setpoint — burner stops at T >= SP	**70**	°C	40 – 90	Client sheet (adjustable from SCADA/HMI)
HYST	Restart hysteresis — burner restarts at T <= SP − HYST	**5**	°C	3 – 10	PH proposal: > type-K accuracy (±2.2 °C), limits short-cycling
STG	2nd stage / high fire while T < SP − STG	**10**	°C	5 – 30	PH proposal
T_START	Max. time from heat request to RUN feedback	**120**	s	90 – 300	PH proposal: ~2 × burner start cycle
T_MAXRUN	Forced stop of BR01 after continuous run	24	h	fixed	RS100 manual §5.10
TDRN_ON	Scrubber high level must last this long before the drain opens (C04)	**3**	s	1 – 30	PH decision 02-Nov-2026: float switch bounce
TDRN_OFF	Drain closes when high level has been gone this long (C04)	**5**	s	0 – 60	PH decision 02-Nov-2026
TDRN_MAX	Drain open continuously longer than this: drain closes, alarm (C04a)	**10**	min	1 – 60	PH decision 02-Nov-2026: no scrubber low level switch

Blue values are defaults; they can be changed from the HMI by an authorised user.

R-11 / D6 / 0007 — PH proposal, accepted internally 01-Nov-2026: lower SP, higher HYST and higher STG apply immediately. Higher SP, lower HYST and lower STG wait until the burner is not Enabled (after STOP or before START); changing a number must not light a burner. All four burner parameters are taken while not Enabled. T_START is also taken while not Active. PLC rejects out-of-range entries, keeps the last valid value and raises an alarm. After power-up, an invalid stored value is replaced by its default.

I/O LIST (from drawing PH-LCP-01 V1, sheets 10–13)

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Address	Tag	Description	Terminal	Signal	Sheet	PLC Tag
I0.0	PSH-101	Scrubber high pressure	XI:1	DI 24V	10	PSH_101
I0.1	LSH-102	Scrubber high level	XI:3	DI 24V	10	LSH_102
I0.2	LSHH-103	Scrubber high-high level	XI:5	DI 24V	10	LSHH_103
I0.3	RR	BR01 run feedback	XI:7	DI 24V	10	BR01_Run
I0.4	S1	BR01 START push button	XI:9	DI 24V	10	S1_BR01_Start
I0.5	Rflt	BR01 fault (lockout)	XI:11	DI 24V	10	BR01_Fault
I0.6	S3	BR01 STOP push button	XI:13	DI 24V	10	S3_BR01_Stop
I0.7	—	Spare	XI:15	DI 24V	10	
I1.0	S0	E-STOP status (NC1)	XI:17	DI 24V	10	S0_EStopStatus
I1.1	S2	FAULT RESET push button	XI:19	DI 24V	10	S2_FaultReset
I1.2	LSL-TK01	TK01 low level	XI:21	DI 24V	10	LSL_TK01
I1.3	LSL-TK02	TK02 low level	XI:23	DI 24V	10	LSL_TK02
I1.4	RR2	BR02 run feedback	XI:25	DI 24V	10	BR02_Run
I1.5	Rflt2	BR02 fault (lockout)	XI:27	DI 24V	10	BR02_Fault
I12.0	Rfuel2	BR02 diesel mode	XI2:1	DI 24V	11	Rfuel2_BR02_DieselMode
I12.1	S4	BR01 AUTO/MAN selector (closed = AUTO)	XI2:3	DI 24V	11	S4_BR01_Auto
I12.2	S5	BR02 AUTO/MAN selector (closed = AUTO)	XI2:5	DI 24V	11	S5_BR02_Auto
I12.3	S6	BR02 START push button	XI2:7	DI 24V	11	S6_BR02_Start
I12.4	S7	BR02 STOP push button	XI2:9	DI 24V	11	S7_BR02_Stop
I12.5	—	Spare (scrubber has no LSL)	XI2:11	DI 24V	11	
I12.6–7	—	Spare	XI2:13–16	DI 24V	11	
Q0.0	R1	BR01 heat request (TL) + H1 1st stage lamp	XQ:1/2	DQ 24V	12	BR01_TL_Cmd
Q0.1	R2	BR01 2nd stage (TR) + H2 2nd stage lamp	XQ:3/4	DQ 24V	12	BR01_TR_Cmd
Q0.2	R3	XV-401 Flare SV (NO)	XQ:5/6	DQ 24V	12	XV_401_Cmd
Q0.3	R4	XV-402 Drain SV (NC)	XQ:7/8	DQ 24V	12	XV_402_Cmd
Q0.4	R5	XV-403 Suction SV (NC)	XQ:9/10	DQ 24V	12	XV_403_Cmd
Q0.5	R6	BR02 start (L)	X6:2	DQ 24V	12 / 8	BR02_L_Cmd
Q0.6	R7	BR02 high fire (T2)	X6:4	DQ 24V	12 / 8	BR02_T2_Cmd
Q0.7	R8	BR02 1st stage lamp H5	—	DQ 24V	12	BR02_Stage1Lamp_Cmd
Q1.0	R9	BR02 2nd stage lamp H6	—	DQ 24V	12	BR02_Stage2Lamp_Cmd
Q1.1	R10	Spare	XQ:19/20	DQ 24V	12	
IW96	TE-501	TK01 inlet oil temperature	X5:1/2	TC type K	13	TE_501_Raw
IW98	TE-502	TK01 outlet oil temperature	X5:3/4	TC type K	13	TE_502_Raw
IW100	TE-503	TK02 inlet oil temperature	X5:5/6	TC type K	13	TE_503_Raw
IW102	TE-504	TK02 outlet oil temperature	X5:7/8	TC type K	13	TE_504_Raw
IW104–110	—	Spare thermocouple channels	X5:9–16	TC	13	